

LESSON 1

MEASURES OF CENTRAL TENDENCY

Overview:

The lesson begins with students engaging in a review of various measures of central tendency. Following the review, students are given cases where these measures are calculated. Students are also asked to examine both strengths and limitations of these measures. Sometime will be devoted to having students discuss questions with a partner before reporting to the class. Assessments will be given to students on their ability to calculate these measures, and also to get an overall sense of whether they recognize how these measures respond to changes in data values.

Learning Outcomes:

At the end of Lesson 1, the students can:

1. calculate commonly used measures of central tendency;
2. provide a sound interpretation of these summary measures; and
3. discuss the limitations of these measures.

Materials Needed:

- Illustrations
- Preliminary Activity Sheet
- Learning Module
- Calculator

Duration: 3 hours

Learning Content:

Measures of Central Tendency

The measures of central tendency are statistical values which describe the point of concentration of many values given in the distribution.

The most commonly used measures of position or central tendency are the **mean**, **median**, and **mode**. The meaning of each measure and how each measure is computed from **ungrouped** (data presented at random or in an array) or from **grouped** (data presented in a frequency distribution) are discussed as follows.

1.1 THE MEAN

The Mean of an Ungrouped Data

The **mean**, also known as the arithmetic average, is found by adding the values of the data and dividing by the total number of values. For example, the mean of 3, 2, 6, 5, and 4 is found by adding $3 + 2 + 6 + 5 + 4 = 20$ and dividing by 5; hence, the mean of the data is $20 \div 5 = 4$. The values of the data are represented by X's. In this data set, $X_1 = 3, X_2 = 2, X_3 = 6, X_4 = 5$, and $X_5 = 4$. To show a sum of the total X values, the symbol Σ (the capital Greek letter sigma) is used, and ΣX means to find the sum of the X values in the data set.

The mean is the sum of the values, divided by the total number of values. The symbol $\bar{X}$ represents the sample mean.

$$\bar{X} = \frac{X_1 + X_2 + X_3 + \cdots + X_n}{n} = \frac{\Sigma X}{n}$$

where n represents the total number of values in the sample.

$$\mu = \frac{X_1 + X_2 + X_3 + \cdots + X_N}{N} = \frac{\Sigma X}{N}$$

where N represents the total number of values in the population.

Example

- 1. The data represent the number of days off per year for a sample of individuals selected from nine different countries. Find the mean. (Source: World Tourism Organization)

20,26,40,36,23,42,35,24,30

Solution

$$\bar{X} = \frac{\Sigma X}{n} = \frac{20 + 26 + 40 + 36 + 23 + 42 + 35 + 24 + 30}{9} = \frac{276}{9} = 30.7 = 31 \text{ days}$$

Hence, the mean of the number of days off is 30.7 days.

Weighted Mean

Mean = $\frac{\text{sum of the products of values and weights}}{\text{sum of the weight}}$ or $\bar{X} = \frac{\Sigma xw}{w}$

Where, Σ is the summation symbol (sigma)

x are values

w are corresponding weights of values

Example

- 2. Find the mean score given that three students get score of 12, two students get score of 15, five students get score of 10, and two students get score of 14.

Solution

- Step 1. Set up columns of weight (*w*) and corresponding values (*x*).
- Step 2. Find the product of *x* and *w* per row.
- Step 3. Find the sum of *w* column and *xw* column

Step 1		Step 2
w	x	xw
3	12	36
2	15	30
5	10	50
2	14	28
12		$\sum xw = 144$

←Step3

- Step 4. Substitute the values in the formula.

$$\bar{X} = \frac{\sum xw}{w} = \frac{144}{12} = 12$$

The Mean of a Grouped Data

Methods:

- 1. Using the Midpoint Method
- 2. Using the Unit – Deviation Method

Method 1: Midpoint Method

Mean = $\frac{\text{sum of products of frequencies and midpoints}}{\text{total number of values}}$ or $\bar{X} = \frac{\sum fmp}{n}$

Where, Σ is the summation symbol read as sigma

f is frequency per class

mp is class midpoint

Steps:

- 1. Set up the class midpoint (*mp*) column.
- 2. Multiply frequency of each class (*f*) and corresponding class midpoint (*mp*).
- 3. Add the results in step 2.
- 4. Add frequency column to get *n*.
- 5. Substitute data in the formula. Solve.

Example

3. Find the mean of the given data below.

Scores	No. of Students <i>f</i>
27 – 29	3
24 – 26	5
21 – 23	12
18 – 20	15
15 – 17	10
12 – 14	8
9 – 11	5
6 – 8	<u>2</u>

Solution

		Step 1	Step 2
Scores	No. of Students <i>f</i>	<i>mp</i>	<i>fmp</i>
27 – 29	3	28	84
24 – 26	5	25	125
21 – 23	12	22	264
18 – 20	15	19	285
15 – 17	10	16	160
12 – 14	8	13	104
9 – 11	5	10	50
6 – 8	<u>2</u>	7	<u>14</u>
$n = 60$		← Step 4	$\sum fmp = 1,086$ ← Step 4

Step 5. Substitute the values in the formula.

$$\bar{X} = \frac{\sum fmp}{n} = \frac{1,086}{60} = 18$$

Method 2: The Unit - Deviation Method

$$\bar{X} = \bar{X}_a + (\frac{\sum fd}{n})i$$

Where, $\bar{X}_a$ is the assumed mean, that is the midpoint of the class claimed as the origin and assigned with 0 deviation.

f is the absolute frequency per class.
 d is the class deviation.
 n is the total number of values.
 i is the interval size, (difference between two consecutive lower limits or upper limits of the class)

Steps

1. Assigned an assumed mean from any one of the midpoints.
2. Set up the unit deviation column by writing 0 opposite the assumed mean, $-1, -2$ and so on, for the succeeding midpoint lower than the assumed mean, and $+1, +2$ and so on, for the succeeding midpoint higher than the assumed mean.
3. Multiply algebraically the deviation of each class by the frequency of the class.
4. Find the values for $\bar{X}_a, \sum fd, n,$ and i and substitute the values in the formula. Solve.

Example

4. Find the mean age using the unit – deviation method.

Ages	No. of Employees
20 – 24	5
25 – 29	9
30 – 34	14
25 – 39	16
40 – 44	11
45 – 49	19
50 – 54	8
55 – 59	3
60 – 64	5

Solution

Ages	<i>f</i>	Step 1 & 2	Step 3
		<i>d</i>	<i>fd</i>
20 – 24	5	–5	–25
25 – 29	9	–4	–36
30 – 34	14	–3	–42
35 – 39	16	–2	–32
40 – 44	11	–1	–11
45 – 49	19	0	0
50 – 54	8	1	8
55 – 59	3	2	6
60 – 64	5	3	15
<i>n</i> = 90		←Step 4	$\sum fd = -117$ ←Step 4

Step 4. $\bar{X}_a = \frac{45+49}{2} = 47, \sum fd = -117, n = 90, i = 20 - 25 = 5$

Step 5. $\bar{X} = 47 + \left(\frac{-117}{90}\right) 5 = 40.5$ years

1.2 THE MEDIAN

The **median** is the midpoint of the ordered data values. The symbol for the median is MD.

Steps in Computing the Median of an Ungrouped Data Set

- Step 1: Arrange the data in order.
- Step 2: If *n* is odd, the middle value is the median of the data set. Otherwise, if *n* is even, the median is the mean of the two middle values of the data set.

Example

Find the median of each of the following data sets.

5. 4,8,1,14,9,21,12

6. 46,23,92,89,77,108

Solution

5. 4, 8, 1, 14, 9, 21, 12

Step 1: Arrange the data in order.

1, 4, 8, 9, 12, 14, 21

Step 2: *n* = 7 and 7 is odd.

1, 4, 8, **9**, 12, 14, 21
 ↑

Median

The middle value is 9. Hence MD = 9.

Solution

6. 46, 23, 92, 89, 77, 108

Step 1: Arrange the data in order.

23, 46, 77, 89, 92, 108

Step 2: $n = 6$ and 6 is even. Thus the median is found halfway between 77 and 89.

23, 46, 77, 89, 92, 108

↑

Median

Hence $MD = \frac{77+89}{2} = 83$.

The Median of a Grouped Data Set

$$MD = lb_{md} + \left[\frac{\frac{n}{2} - cf_{<md}}{f_{md}} \right] i$$

Where, lb_{md} is the lower boundary of the median class.

n is the total cases in the distribution.

i is the interval size.

$cf_{<md}$ is the less cumulative frequency entry before the Median class.

md is the class with $cf_{<md}$ entry equal to or slightly higher than $\frac{n}{2}$.

Steps:

Step 1. Find $\frac{n}{2}$.

Step 2. Locate the median class by accumulating frequencies by the less than method in to a value $\frac{n}{2}$ or slightly higher than $\frac{n}{2}$.

Step 3. Find values corresponding to lb_{md} , $cf_{<md}$, f_{md} , and i .

Step 4. Substitute data in the formula and solve.

Example

7. The following table shows the frequency distribution of ages of 90 employees in a government agency. Find the median.

Ages	No. of Employees
20 – 24	5
25 – 29	9
30 – 34	14
35 – 39	16
40 – 44	11
45 – 49	19
50 – 54	8
55 – 59	3
60 – 64	5

Solution

Step 1: $\frac{n}{2} = \frac{90}{2} =$

45

Step 3:

$lb_{md} = \frac{39+40}{2} = 39.5$

$cf <_{bmd} = 44$

$f_{md} = 11$

$i = 5$

Median Class→

Step 2

Ages	<i>f</i>	<i>cf</i> <
20 – 24	5	5
25 – 29	9	14
30 – 34	14	28
35 – 39	16	44
40 – 44	11	55
45 – 49	19	74
50 – 54	8	82
55 – 59	3	85
60 – 64	5	90
<i>n</i> = 90		

$\leftarrow cf <_{bmd}$

$\leftarrow f_{md}$

Step4:

$$MD = lb_{md} + \left[\frac{\frac{n}{2} - cf <_{bmd}}{f_{md}} \right] i$$

$$MD = 39.5 + \left(\frac{45 - 44}{11} \right) 5$$

$$MD = 39.5 + \left(\frac{1}{11} \right) 5$$

$$MD = 39.5 + \left(\frac{5}{11} \right)$$

$$MD = 39.5 + 0.45$$

$$MD = 39.95 \text{ years or } 40 \text{ years}$$

Thus out of 90 employees, about 45 of them have ages below 40 years while the other 45 employees have ages equal to or higher than 40 years.

1.3 THE MODE

The mode is the value that occurs most often in the data set. It is sometimes said to be the most typical case.

A data set that has only one value that occurs with the greatest frequency is said to be **unimodal**. If a data set has two values that occur with the same greatest frequency, both values are considered to be the mode and the data set is said to be **bimodal**. If a data set has more than two values that occur with the same greatest frequency, each value is used as the mode, and the data set is said to be **multimodal**. When no data value occurs more than once, the data set is said to have **no mode**.

Mode of Ungrouped Data

Example

Find the mode of the data in the following lists.

8. 18, 15, 21, 16, 15, 14, 15, 21 9. 2, 5, 8, 9, 11, 4, 7, 23

Solution

8. In the list 18, 15, 21, 16, 15, 14, 15, 21, the number 15 occurs more often than the others. Thus 15 is the mode.
9. Each value in the list 2, 5, 8, 9, 11, 4, 7, 23 occurs only once. Thus, there is no mode.

Mode of Grouped Data

Methods:

Method 1. Estimating the mode by the rough method:

mo = midpoint of a class with the highest frequency

Example

10. Using the data from example 7, the mode of the ages of the 90 employees in a government agency is the midpoint of the class interval 45 – 49, which is 47 years.

Method 2. Using the formula

$$mo = lb_{mo} + \left(\frac{d_1}{d_1 + d_2} \right) i$$

Where, lb_{mo} is the lower boundary of the modal class.

d_1 is the difference between the frequency of the modal class

and the frequency of class immediate lower than the modal class.

d_2 is the difference between the frequency of the modal class and the frequency of class immediate higher than the modal class.

i is the interval size.

Steps:

- Step 1. Find the modal class.
- Step 2. Find the values corresponding to lb_{mo} , d_1 , d_2 , and i .
- Step 3. Substitute the values in the formula and solve.

Example

11.Find the mode of the frequency distribution showing the age of 90 employees in a government agency from example 7.

Solution

		Steps:	
		1. Modal Class is 45 – 49.	
		2. $lb_{mo} = \frac{44+45}{2} = 44.5$	
		$d_1 = 19 - 11 = 8$	
		$d_2 = 19 - 8 = 11$	
		$i = 5$	
		3. $mo = lb_{mo} + \left(\frac{d_1}{d_1+d_2}\right)i$	
		$mo = 44.5 + \left(\frac{8}{8+11}\right)5$	
		$mo = 44.5 + \left(\frac{8}{19}\right)5$	
		$mo = 44.5 + \left(\frac{40}{19}\right)$	
		$mo = 44.5 + 2.1$	
		$mo = 46.6$	
		$mo = 47 \text{ years}$	

		Ages	No. of Employees
		20 – 24	5
		25 – 29	9
		30 – 34	14
		35 – 39	16
		40 – 44	11
Modal Class→		45 – 49	19
		50 – 54	8
		55 – 59	3
		60 – 64	5

Learning Activities:

Activity 1

Direction: Through Google Form, students are requested to provide the necessary information (i.e. age, height, weight, number of siblings, monthly internet allowance, general mathematics final grade, and statistics and probability final grade) prior to this activity. Afterwards, students are divided into seven (7) groups to find the **mean**, **median**, and **mode** of the following variables and provide a sound interpretation of the resulting values.

- Group 1:** Age
- Group 2:** Height
- Group 3:** Weight
- Group 4:** Monthly Expenses
- Group 5:** Monthly Internet Allowance
- Group 6:** General Mathematics Final Grade
- Group 7:** Statistics and Probability Final Grade

Activity 2

Direction: Compute for the **mean**, **median**, and **mode** of the following grouped data. The same group as *Activity 1.1* will work in this activity. Give an interpretation in each resulting measure.

GROUP 1		GROUP 2		GROUP 3		GROUP 4	
Age	f	Weight (in kg)	f	Daily Expenses (in pesos)	f	Number of Hours Spent in Social Media in a Week	f
10 - 14	3	110 - 119	6	200 - 249	13	100 - 109	2
15 - 19	19	120 - 129	14	250 - 299	11	90 - 99	5
20 - 24	11	130 - 139	5	300 - 349	16	80 - 89	14
25 - 29	10	140 - 149	15	350 - 399	14	70 - 79	15
30 - 34	2	150 - 159	8	400 - 449	7	60 - 69	6
35 - 39	5	160 - 169	7	450 - 499	9	50 - 59	3

GROUP 5		GROUP 6		GROUP 7	
Score	f	Monthly Income (in pesos)	f	Cash in Bank (in pesos)	f
0 - 3	4	3,000 - 11,999	2	100 - 249	1
4 - 7	16	12,000 - 23,999	12	250 - 299	5
8 - 11	12	24,000 - 35,999	13	300 - 349	24
12 - 15	8	36,000 - 47,999	23	350 - 399	6
16 - 19	10	48,000 - 59,000	6	400 - 449	3
20 - 23	5	60,000 - 71,000	4	450 - 499	1

Learning Evaluation:

1. The following table displays the mothers' ages of the 30 GEC 104 students. Find the mean, median and mode of the data set.

36	44	47	52	45
40	39	51	58	51
39	42	50	36	38
37	51	38	55	35
45	38	43	47	44
46	50	34	48	41

2. After 6 Biology tests, Ruben has a mean score of 78. What score does Ruben need to raise his average to 80?
3. An article recently reported that the median income for college professors was ₱43,250. What does this mean?
4. If the set of data has a mode, then must the mode be one of the data values? Explain.

References:

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Blay, B. (2013). *Elementary Statistics*. Anvil Publishing, Inc. Mandaluyong City, Philippines.